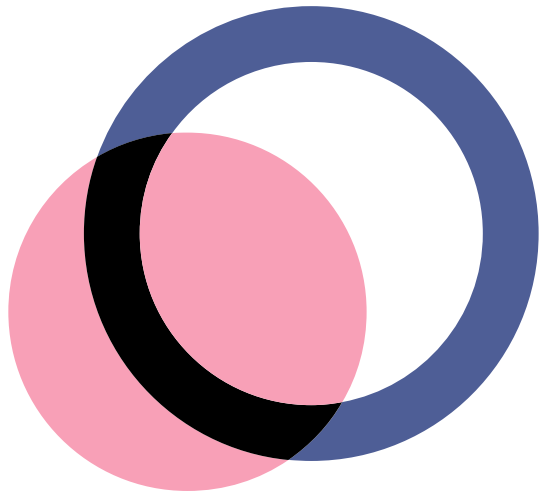
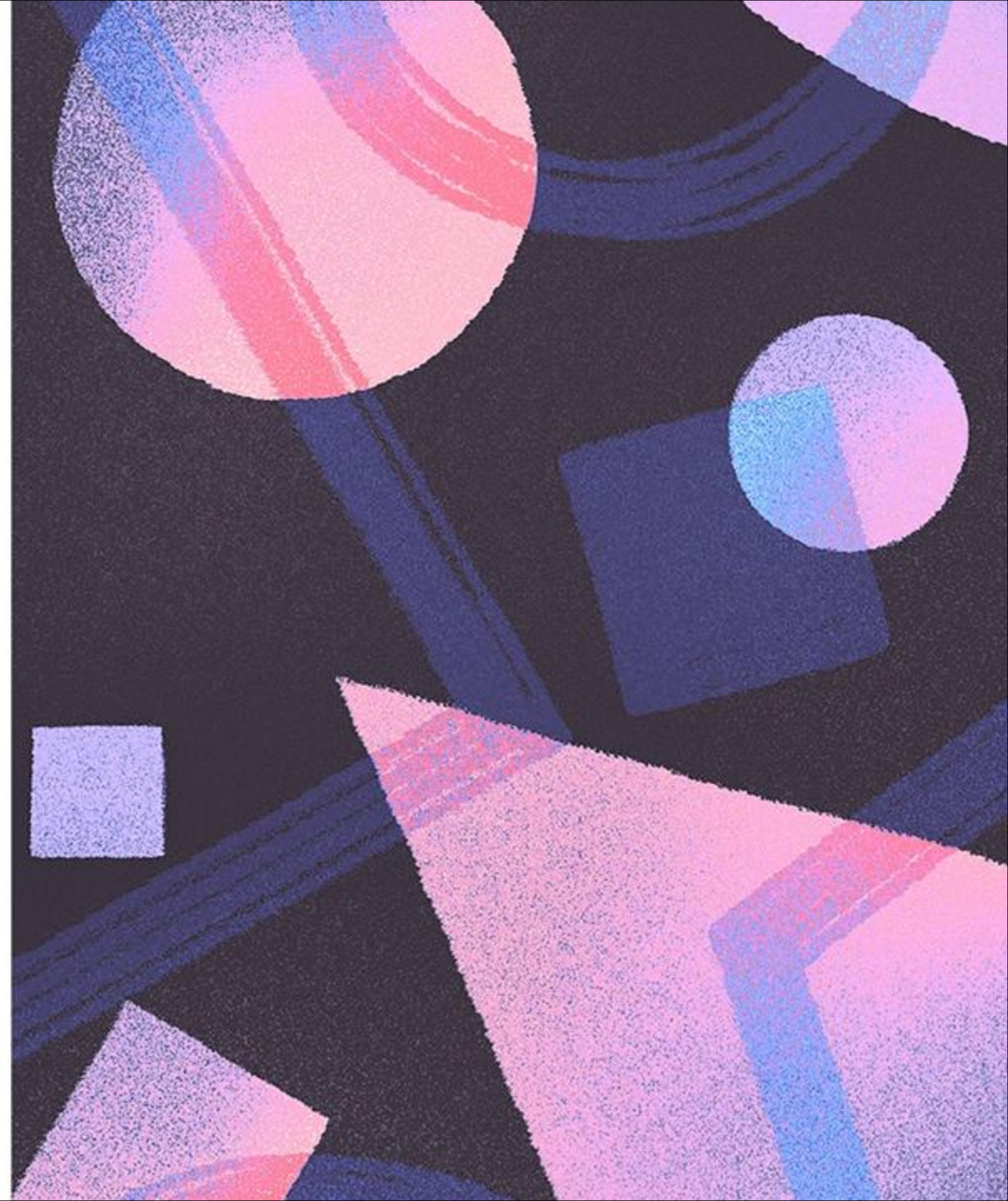




Data Structures Lab

Lecture 6

> Trees



CONTENT

01

PART

Trees

02

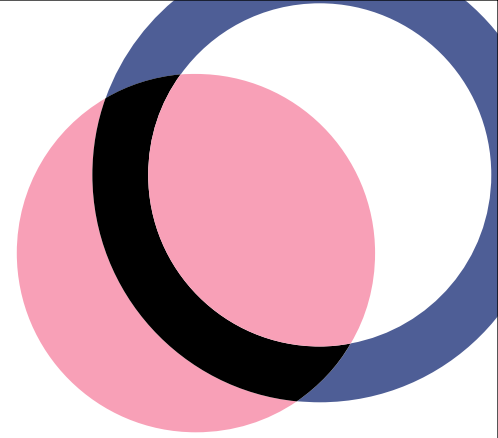
PART

Binary Trees

03

PART

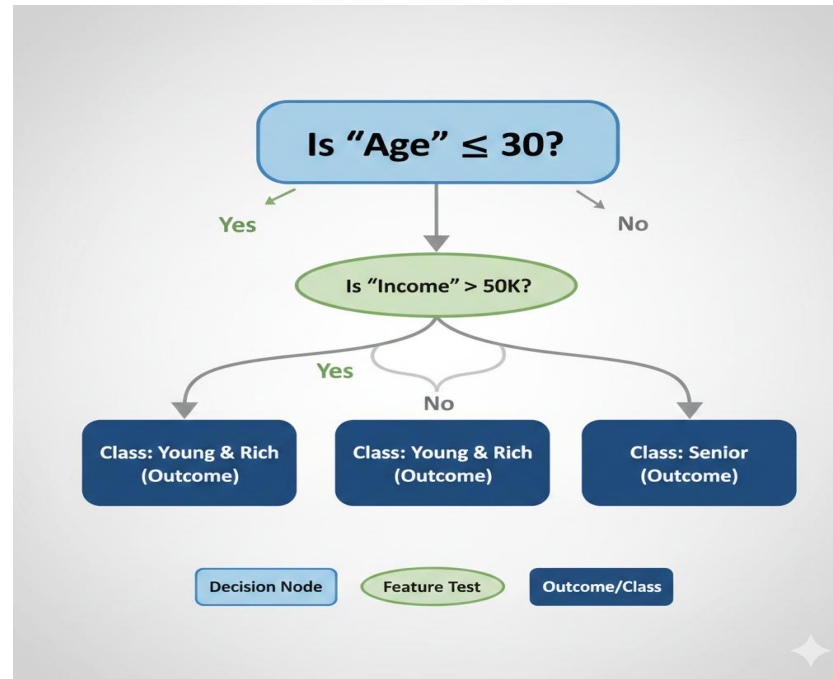
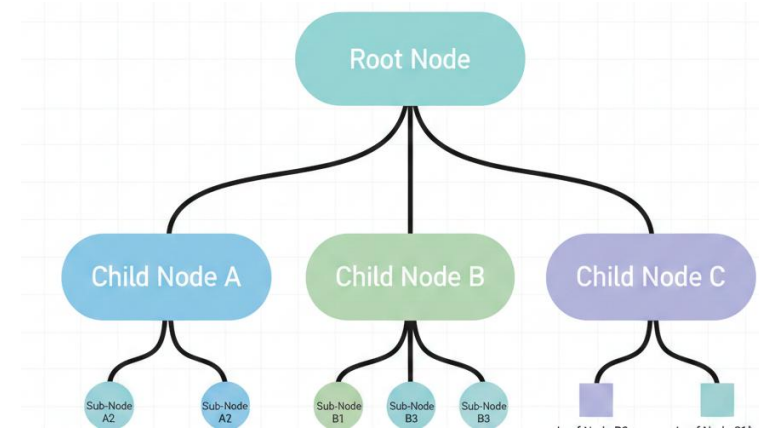
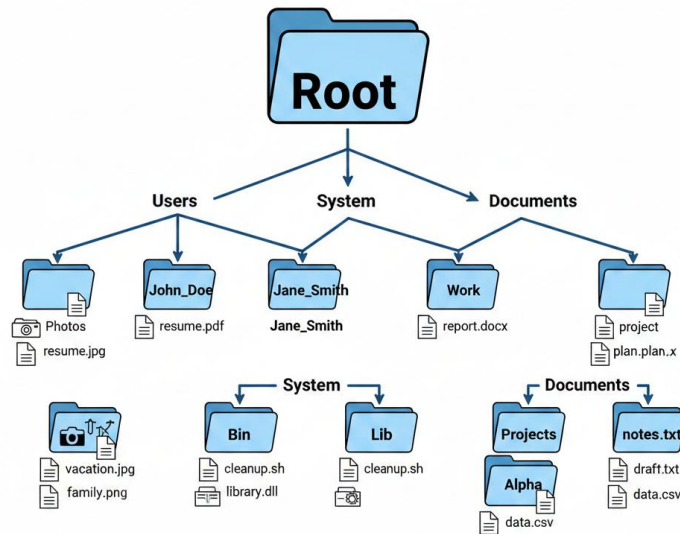
Examples



01: Trees: A Non-Linear Data Structure

- Non-linear structures
- Hierarchical data representation
- Applications in file systems, decision making, and more

File System Tree Structure



What is a Tree?

- A tree is a collection of nodes connected by edges.
- Has a root node, internal nodes, and leaf nodes.
- Hierarchical structure with parent-child relationships.

Linear vs Non-Linear Data Structures

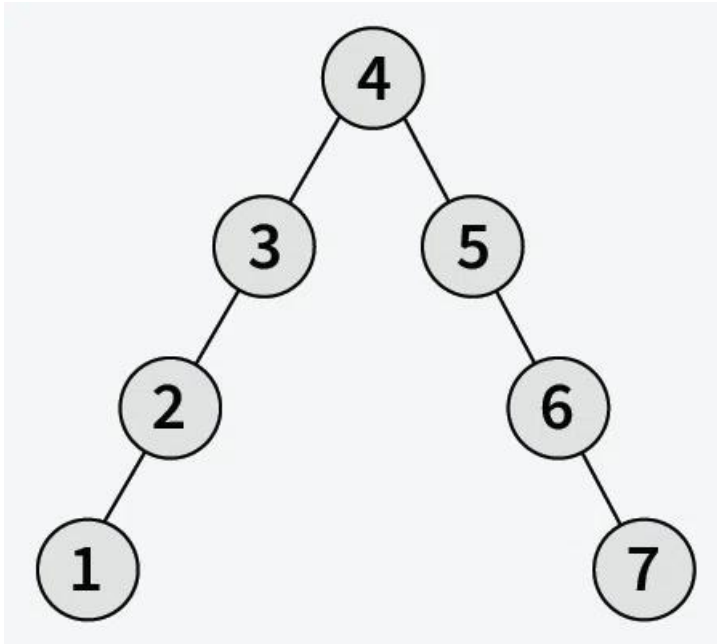
- Linear structures: Arrays, Linked Lists, Stacks, Queues
- Non-linear structure: Trees
- Trees allow for branching, unlike linear structures.

Tree Terminology

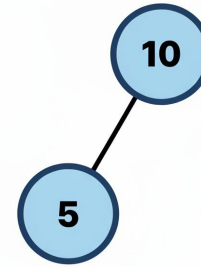
- Root: The starting node.
- Node: Any element in the tree.
- Child/Parent: Nodes connected by edges.
- Leaf: A node with no children.

Binary Trees

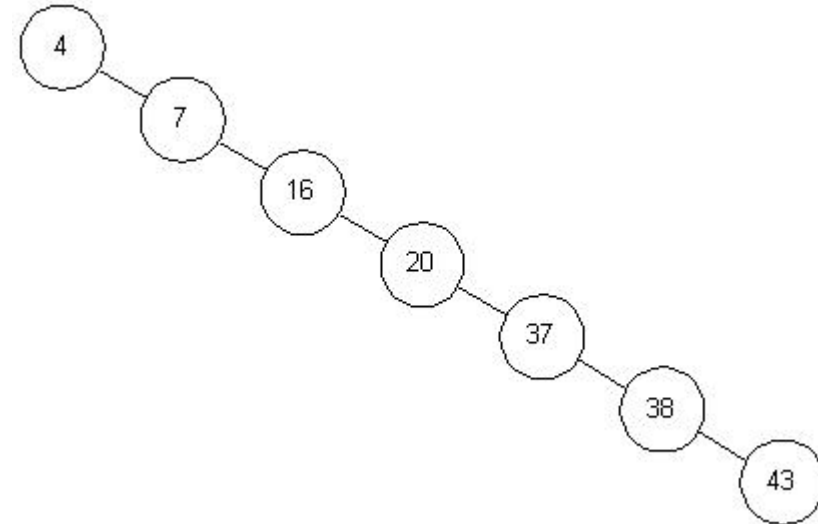
- Each node has at most two children: left and right.
- Used in various applications like search.



Binary Tree



BINARY TREE



Tree Traversals

1. Depth-First Search (DFS)

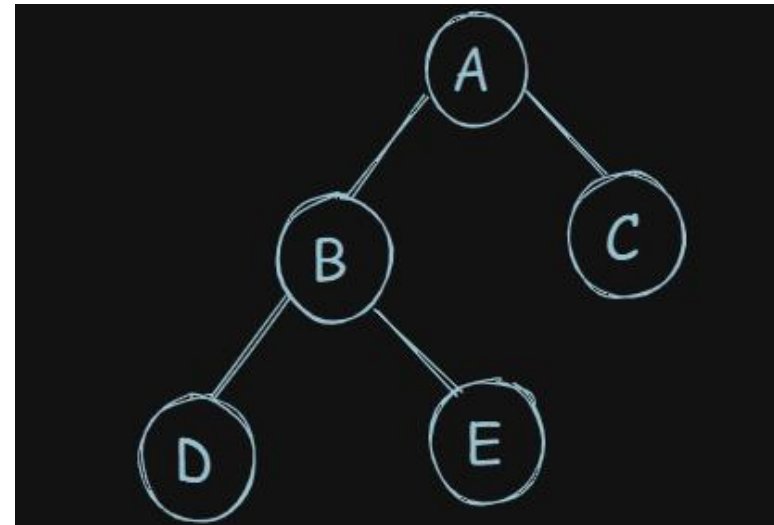
- DFS explores as deep as possible along a branch before backtracking.
- In trees, DFS has three standard traversal orders.

1.1 Preorder Traversal (Root → Left → Right)

Steps:

1. Visit the root
2. Traverse the left subtree
3. Traverse the right subtree

Traversal Order: **A** → **B** → **D** → **E** → **C**



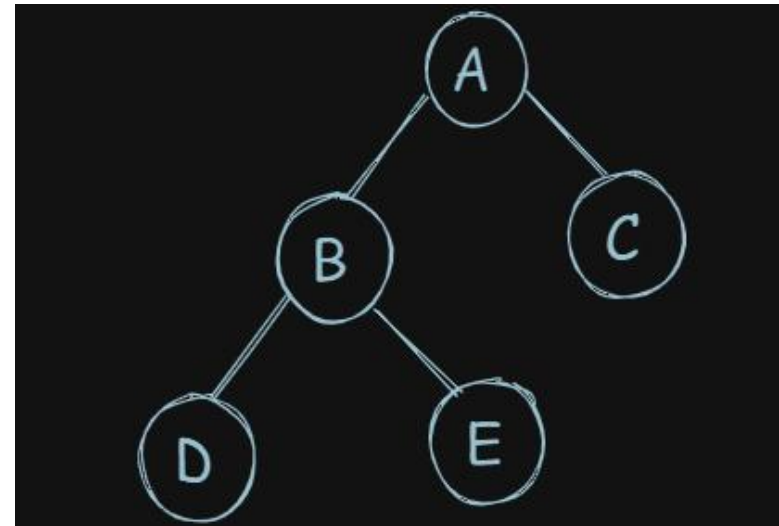
Tree Traversals

1.2 Inorder Traversal (Left $\rightarrow$ Root $\rightarrow$ Right)

Steps:

1. Traverse the left subtree
2. Visit the root
3. Traverse the right subtree

Traversal Order: **D $\rightarrow$ B $\rightarrow$ E $\rightarrow$ A $\rightarrow$ C**



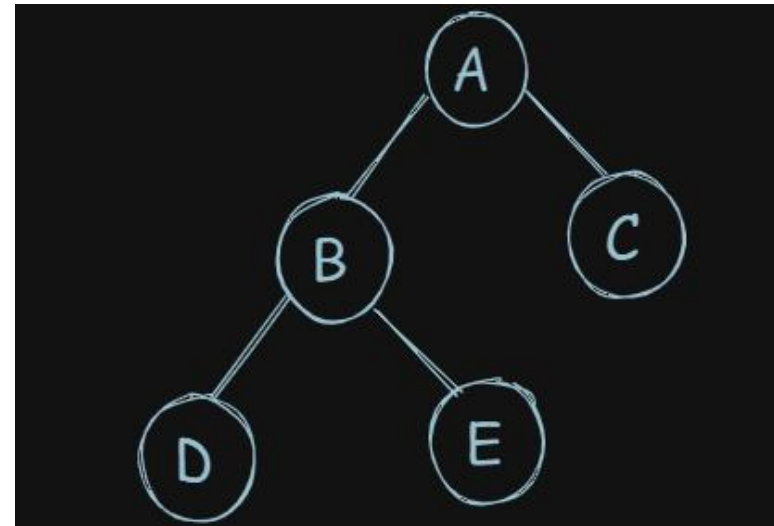
Tree Traversals

1.3 Postorder Traversal (Left $\rightarrow$ Right $\rightarrow$ Root)

Steps

1. Traverse the left subtree
2. Traverse the right subtree
3. Visit the root

Traversal Order: **D** $\rightarrow$ **E** $\rightarrow$ **B** $\rightarrow$ **C** $\rightarrow$ **A**

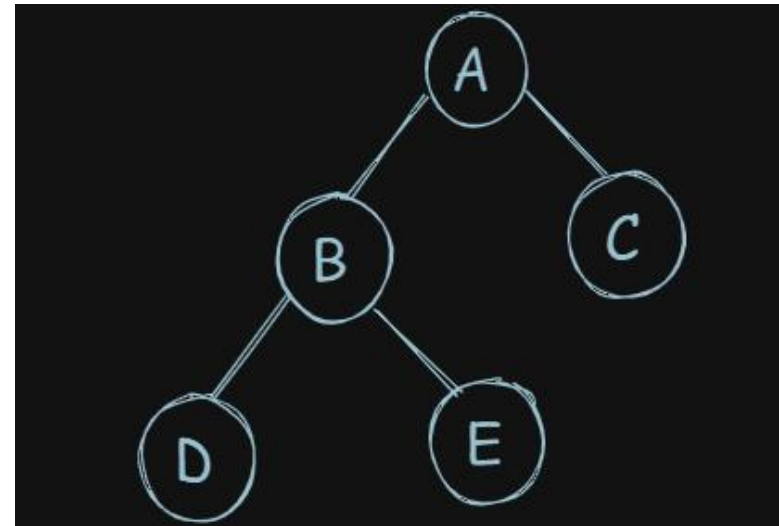


Tree Traversals

2. Breadth-First Search (BFS)

- BFS explores the tree level by level, starting from the root and moving left to right.
- Level Order Traversal (BFS)

Traversal Order: **A** → **B** → **C** → **D** → **E**

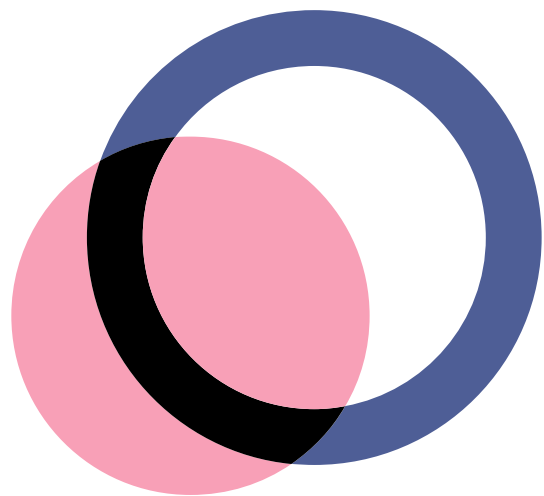


Applications of Trees

- File Systems (folders and files)
- Decision Trees

Summary & Next Steps

- Trees are used to represent hierarchical data.
- They provide an efficient way to search and store complex relationships.
- Next lecture: Binary Search Trees (BST)!



**THANK
YOU**

